

PAPER_1454 — UQFF: DM Direct Detection σ Floor (Bucket K)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket K **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — formal catalog tuple in Bucket K **Calculator surface:** `calculate_bsm_constraints({...})`

Closure

$\Lambda^4 \times 1e-40 \text{ cm}^2$

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_bsm_constraints_report()` or equivalent surface in `uqff_pure_calculator.py`
 - Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.
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